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# Robust method for road sign detection and recognition

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**Abstract**

This paper describes a method for detecting and recognizing road signs in grey-level and colour images acquired by a single camera mounted on a moving vehicle. The method works in three stages. First, the search for the road sign is reduced to a suitable region of the image by using some *a priori* knowledge on the scene or colour clues (when available). Secondly, a geometrical analysis of the edges extracted from the image is carried out, which generates candidates to be circular and triangular signs. Thirdly, a recognition stage tests by cross-correlation techniques each candidate which, if validated, is classified according to the database of signs. An extensive experimentation has shown that the method is robust against low-level noise corrupting edge detection and contour following, and works for images of cluttered urban streets as well as country roads and highways. A further improvement on the detection and recognition scheme has been obtained by means of temporal integration based on Kalman filtering methods of the extracted information. The proposed approach can be very helpful for the development of a system for driving assistance.

*Keywords:* Traffic automation; Shape detection; Road sign recognition

## 1. Introduction

Many applications of vision systems to traffic automation have recently been developed [1–8]. These applications include traffic control and analysis, license plate finding and reading, toll collection, automatic route planning and passive navigation. The automatic detection and classification of road signs clearly belongs to this emerging field of applications [7,9–12]. This paper describes a visual system for the detection and recognition of road signs. It shows that common low-level techniques like edge detection and template-matching, along with geometrical reasoning based on some a priori knowledge on the environment and on the road signs, can be used successfully to provide the driver with relevant information on the scene. This is achieved by a three-stages algorithm which, after reducing the search to a suitable area of interest, looks for contours of triangular or circular shape by analysing the edges extracted from a single monochromatic or colour image. Whenever a candidate to be a road sign is found, the recognition procedure is activated, which has the twofold purpose of validating the candidate and possibly classifying it.

Since outdoor working conditions can vary considerably, special attention has been devoted to the robustness and flexibility of the system: the low-level routines have been designed effectively to handle both noisy input data and complex environments such as cluttered urban scenes. Such distinctive features are exalted by a temporal integration based on Kalman filtering methods of the information extracted from a single frame.

A wide experimentation on hundreds of images taken by a vehicle moving in urban environments, country roads and highways, indicates that the proposed approach for the detection and recognition of road signs is indeed effective and useful for the realization of driving assistance module apt to be used in different weather conditions and environments.

### *1.1. Outline of the paper*

Section 2 describes the experimental set-up and the assumptions that are made on the scene and on the image. The core of the paper is Section 3, where the detection and recognition algorithm is described. Section 4 addresses the temporal integration of the extracted information.

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## 2. Experimental set-up and assumptions

The images that will be analysed in this paper have been acquired at video-rate by a camera mounted on a vehicle (a van) at the height of about 1.70 m. The optical axis of the camera was roughly parallel to the roadbed, and oriented in the direction of motion. Assume that a cartesian system of reference $(O; X, Y, Z)$ solid with the imaging device is given, so that the origin of the reference system $O$ coincides with the optical centre and the motion is along the $Z$-axis. The image plane is defined by the equation $Z = f$. The generic 3D point $P = (X, Y, Z)$ with $Z \geq f$ is projected into the point

$$\pi(P) = \left( f\frac{X}{Z}, f\frac{Y}{Z}, f \right)$$

on the image plane, where $f$ is the focal length of the imaging device (in our case, $f = 595 \pm 5$ pxl). The vehicle can be assumed to proceed parallel to the road direction. As a consequence, the prolongation of the segments which are projection of the edges of the road (in the stretch between the vehicle and the road sign) converge on the centre of the image, which is the vanishing point associated with the optical axis direction.[1] Assume that the stretch of road between the vehicle and the road sign is straight. The road sign is assumed to be orthogonal to the road direction, and therefore parallel to the image plane, as it usually is in practice. As a consequence, the transformation between the 3D sign and the projected sign is a similarity transformation, so that there is no perspective distortion of the shape of the sign: a circular sign remains a circle and the sides of a triangular sign do not change their slope.

## 3. Detection and recognition algorithm

Following the indications of several authors (see, for example, [11,13]), the detection and recognition of road signs is carried out in three stages. First, a region of the image in which the road sign is more likely to be found is selected, by using either the information on colour or some a priori information on where the sign is supposed to be. The search is then limited to this region, called a search region, or region of interest. Secondly, a geometrical analysis is carried out on the output of an edge detector applied to the search region, and sets of edge-points having the desired shape (e.g. triangular or circular) are extracted. Each set is a candidate to be the contour of a road sign. Thirdly, the inner region of each candidate is tested against the database of signs by template-matching and validated or discarded depending on whether there is an element in the database whose cross-correlation with the candidate is greater than a given threshold. As an output of this stage, the road sign is not only detected but also classified.

### 3.1. Extracting the search region

The amount of computation for detecting the road signs in the image can be greatly reduced if the search is limited to a small region of the image where the road sign is more likely to be found. There are of course several ways to select a suitable search region, depending on the presence of clues like colour or other a priori information on the localization of the road sign in the image. Two simple methods are described: in the first one (Section 3.1.1), only grey-level information is available and the region is chosen according to some simple geometrical criteria; in the second one (Section 3.1.2), the information on colour is used: the region of interest becomes the one in which there is a sufficient number of image elements whose colour is similar to the desired one.

#### 3.1.1. A priori localization

Even when no information on colour is given, the search for road signs can be reduced to a limited portion of the image, by taking into account the a priori knowledge on the localization of the road sign in the scene. By exploiting some simple and fast segmentation technique (see, for example, [3]), the nearly uniform regions of the sky and of the road can be detected, so as to limit the search to the remaining part of the image (the search can be further limited to the right part of the image). It is, however, computationally disadvantageous to handle a search region which is not rectangular. A suitable rectangular region, which has met our needs, is the middle-right part of the image (see Fig. 1).

#### 3.1.2. Localizing by colour

The information on colour provides the quickest way to reduce the search for road signs to a small region of the image, and was used whenever possible. The use of colour for image segmentation corresponds to the way in which road signs are usually spotted by human drivers.

To simplify the description of the segmentation algorithm, the case of triangular (danger) signs is now considered, whose border can only be red. The search for circular signs simply implies the search for a region which can be either blue or red. The algorithm can be outlined as follows:

1. Each image element is classified as 1 (red) or 0 (non-red) according to its hue (which must be between −30 and 30 in an angular scale which assigns 0 to red, 120 to green and 240 to blue) and saturation (which must be at least 20%).

[1] Recall that the vanishing point VP $= (x, y)$ in the image plane $Z = f$ is the meeting point in the image of all 3D edges whose direction is given by the unit vector $\vec{e} = (x, y, f)/\sqrt{x^2 + y^2 + f^2}$.

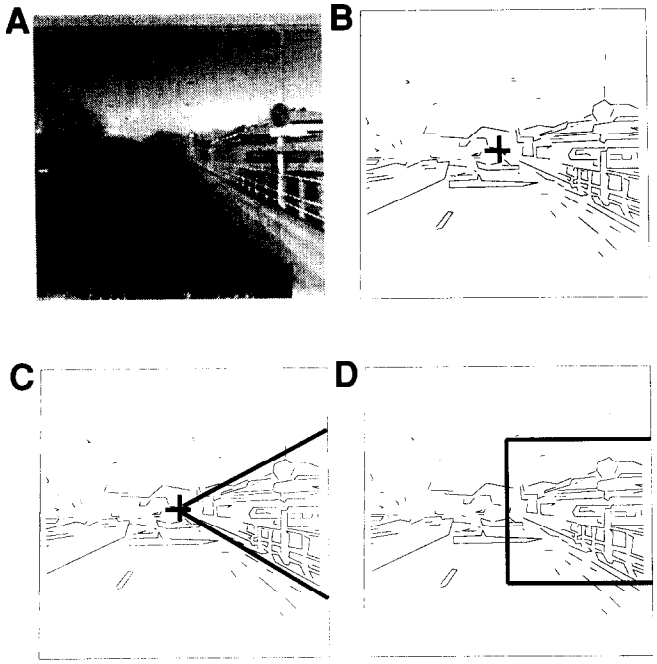



Fig. 1. Segmentation of the image. (a) The original image (512 × 512 pixels, 8 bits deep); (b) central vanishing point. The edges were obtained by Canny algorithm [14] with a Gaussian filter with $\sigma = 1.5$. Edge chaining and polygonal approximation have been performed by Pavlidis algorithm [23]. The central vanishing point has been extracted by the algorithm described in [24]; (c) the search region, which was recovered by the algorithm described in [3] for the detection of road borders and other 3D features in typical road scenes; (d) search region that was actually used.

2. The 512 × 512 image is subdivided into 16 × 16 regions. Each region is classified as 1 or 0 according to whether the number of pixels labelled as 1 exceeds a certain threshold (80, in our case).
3. A search region is associated with every cluster of '1'-regions. A cluster can be defined as a maximal set of pairwise adjacent '1'-regions, as in Fig. 2(a). The outer boundary of these clusters can be simply obtained by means of the usual techniques for extracting the outer boundary of the planar embedding of a graph. Once the outer boundary of the cluster has been determined, the search region can be defined as the 'rectangular hull' of the cluster (that is, the smallest rectangle containing the cluster itself) augmented with a frame of eight pixels (see Fig. 2(b)), which reduces the risk of underestimating the region of interest.[2]

Each region constructed by the algorithm described above is investigated for detecting a road sign. Observe that there may be two regions $R_i$, $R_j$ such that

[2] The choice of a rectangular shape was made on account of the fact that, lacking dedicated hardware, scanning a rectangular mask is the best bet from a computational standpoint; furthermore, extracting an ad hoc shape (a circle or a triangle) is sensible only when it is assumed that the clustering algorithm manages to capture the exact boundary of the road sign, which is too strong an hypothesis in most cases.

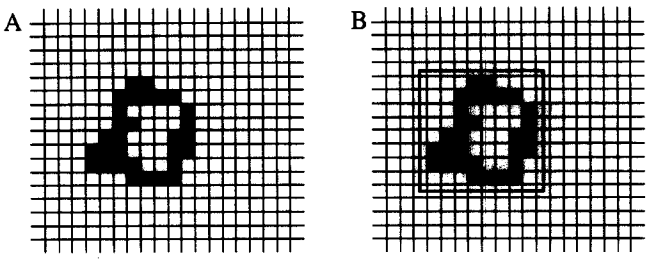



Fig. 2. A simple illustration of the extraction of the search region when colour clues are available. (a) Image subdivided into regions of 16 × 16 pixels. Black squares represent '1'-regions, that is regions where the percentage of red (for example) pixels is greater than a given threshold. To extract the search region, clusters of pairwise adjacent '1'-regions must be identified in the first place. In (a) there is one of such clusters. The search regions (there may be more than one) are defined as the rectangular hulls containing the clusters of '1'-regions, augmented with a frame of 8 pixels; (b) search region associated with the cluster of (a).

$R_i \cap R_j \neq O$. In this case, each region is examined by itself. This is in practice less expensive than merging $R_i$ and $R_j$ to find the rectangular hull comprising both regions (the trivial merge, the one which gives us $R_i \cup R_j$, is obviously not convenient, as we always wish to deal with rectangular regions), in that the area of the rectangular hull of $R_i \cup R_j$ is usually greater (in our examples) than the area of $R_i$ summed up with the area of $R_j$.

An example of the application of this algorithm is shown in Fig. 3. Fig. 3(a) shows the value (brightness) component of a colour image in *HSV*-representation. Fig. 3(b) shows all points of the image which are classified as red according to Step 1 of the algorithm described

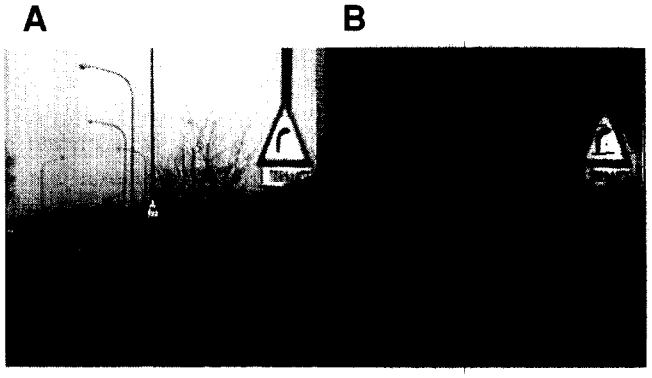



Fig. 3. An example of the application of the extraction of the search region by using colour. (a) Value (brightness) component of a colour image in HSV-representation; (b) value component of all points whose saturation is greater than 20% and whose hue is in the range [−30, 30] (conventionally chosen to represent an acceptable red). Observe that the white region inside the sign is identified as red by our algorithm, due to the fact that the photograph is taken at sunset (see Also Section 5). The search region, obtained from these points by the algorithm described in the text, is superimposed to the image. It consists of two rectangles roughly centred about the road signs. Observe that there are points which are identified as red but do not belong to any search region. This is because no region of 16 × 16 pixels has been found with at least 80 red pixels.

above. The search region is composed by two rectangles which are roughly centred about the road signs.

### 3.2. Shape detection

This section addresses the problem of detecting road signs within the search region extracted as described in Section 3.1. The proposed approach, based on the geometrical analysis of the edge-contours, aims at retrieving circular and triangular contours representing possible candidates to be road signs. For both the circular and triangular shape recovery algorithms, the starting information is represented by the edge map, which is extracted from the search region by Canny's algorithm [14] (see legend of Fig. 4 for numerical details). It is to be noticed that, although our method for shape detection uses the output of an edge detector, it does not rely heavily on the good quality of such an output (e.g. it is not necessary that the contour of the road sign be an unbroken closed curve).

#### 3.2.1. Triangular road signs

Fig. 4 illustrates the basic steps of the algorithm. Figs 4(a–c), respectively, reproduce an image of a road scene, the selected search region, and the related edges.

The proposed procedure can be summarized in the following steps:

1. A polygonal approximation of the edge chains inside the search region (see Fig. 4(d)) is first performed to discard those parts of the chains that strongly depart from a straight segment. This can be achieved by simply thresholding the length of the approximating segments.
2. Under the assumption that the vehicle moves along the road direction and that the road signs are orthogonal to the road direction, the shape of the sign – which is an equilateral triangle – is not distorted by perspection. On account of this, the search can be limited to quasi-equilateral triangles with a horizontal side. The sides of a real triangular sign should therefore be mapped into segments belonging to one of the following sets:
   (a) segments with slope in the range $[-\epsilon, \epsilon]$ degrees with respect to the horizon line;
   (b) segments with slope in the range $[60 - \epsilon, 60 + \epsilon]$ degrees with respect to the horizon line;
   (c) segments with slope in the range $[60 - \epsilon, -60 + \epsilon]$ degrees with respect to the horizon line.

   The parameter $\epsilon$, which takes into account noise effects and possible deviations from the hypothesis of a road sign exactly orthogonal to the image plane, has been experimentally found not to be critical. Fig. 4(e) displays the segments thus selected for the image of Fig. 4(a).
3. Once all segments of proper slope have been selected, it must be checked whether they can be assembled so as to belong to the same equilateral triangle. This can be done as follows. A structure vertex is created for every pair of segments whose endpoints are closer than a given threshold and form an angle of about 60 degrees. The vertices are subdivided into three classes, according to their orientation. The recovery of the triangle, if there is one, is then performed according to one of the following methods. If three vertices belonging to three different classes are found, it is simply checked whether they correspond to a suitable triangle. If some triangles are found by this method, their corresponding segments are eliminated before moving on to the following step, which tries to reconstruct triangles from pairs of segments. If

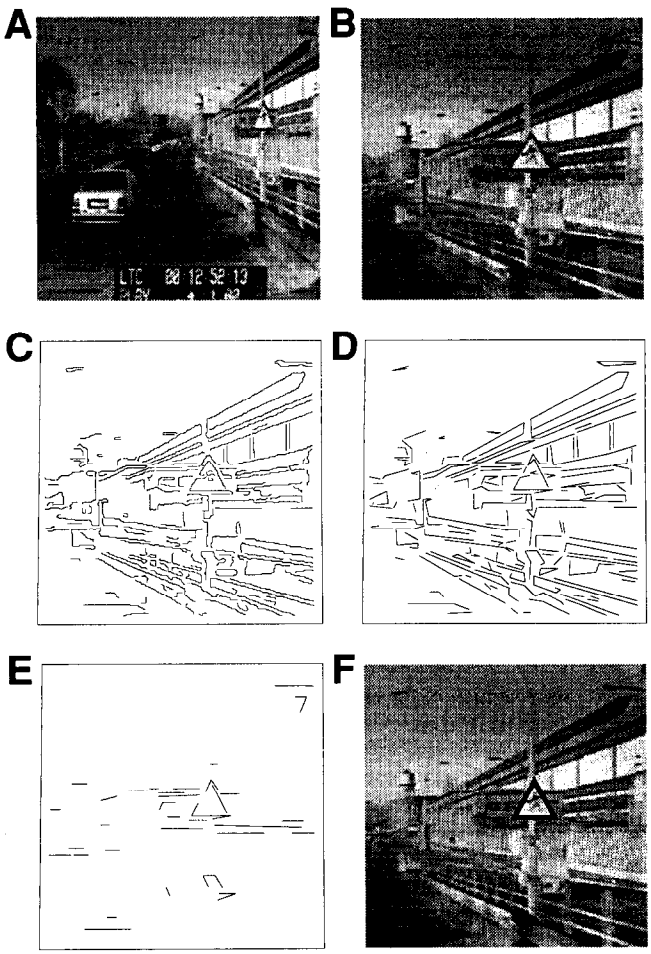



Fig. 4. Recovery of triangular road signs. (a) The original image; (b) search region extracted from (a) according to the segmentation criterion described in Fig. 1; (c) (d) two different stages of the low-level processing of image (b); (c) output of the Canny edge detector. Only edges with a strength (the modulus of the intensity gradient) greater than a certain threshold are considered. The Gaussian filter has $\sigma = 1.5$; (d) polygonal approximation of edges in (c) by the Pavlidis algorithm with minimum length of the segments $l = 2$, and maximum distance of the segment from the chain $d = 2$; (e) segments of proper length and slope are selected and classified in three sets according to their slope. The triangular contour is recovered (f) by properly assembling the segments of (e).

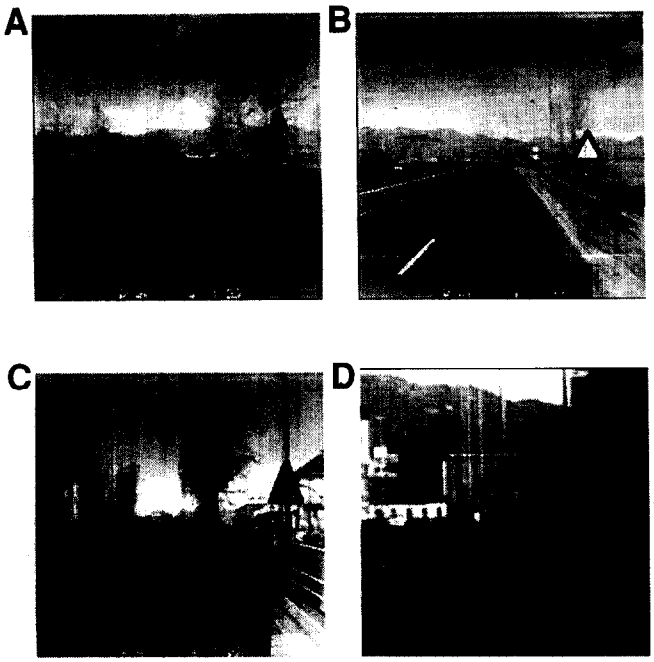



Fig. 5. Examples of triangular road sign recovery. (a)–(c) Examples of successful detection; (d) example of failure, due to the lack of contrast between the triangular road sign and the background.

vertices of only two different classes are found, a consistent reconstruction with two vertices is attempted by analysing the segments which form these vertices. Even if vertices of only one class are found, or if the reconstruction with two vertices has failed, it is possible to try to reconstruct a triangle. In this case, the segments of Step 2 are considered again: the segments belonging to the class with a suitable slope are examined in order to find the missing side of the triangle. If such a segment is found, a triangle is recovered. Observe that the output of this algorithm can consist of several triangles.

The final result of this procedure is shown in Fig. 4(f), where the detected road sign is superimposed onto the original image. The overall procedure, from the grey-level image to the detection of triangles, takes about 6 seconds on a SparcStation ELC. Other examples of the application of the described algorithm to different road scenes can be found in Fig. 5.

The experimentation over 600 images containing one or more triangular signs has given about 92% of correct identifications, and 11 false alarms (i.e. reports of non-existing signs).

*3.2.2. Circular road signs*

The detection of circular road signs is more difficult than the detection of triangular ones. A natural counterpart of the algorithm described in Section 3.2.1 would be to look for chains of edge points of constant curvature, but this leads to very unstable results and a poor rate of success at detection. This section describes the design of a new, more robust algorithm for the detection of circular shapes,[3] and compares it with the other methods that are commonly used for finding circles in an image. The algorithm, which looks for circles of variable radius by inspecting the search region with annuli of variable radius and fixed thickness, can be outlined as follows (see Fig. 6):

1. The edge map (see Fig. 6(c) and (d)) of the search region is extracted.
2. The edge map of the search region is explored by means of annuli of increasing radius and fixed thickness (usually 3 pxl).
   For every position of the annuli in the image, the number of edge points between the internal and external radius is counted: only the positions where the number of edge points exceeds some given threshold are selected.
3. Contour following is performed on the edge map to analyse the edge points selected as in Step 2.
4. Three tests (on the radial and angular distribution of the edges, and the length of the longest connected edge-chain) are now performed on the edge contours contained in the selected annuli. In this way, we can recover also partially corrupted circular contours, compensating for the poor accuracy of the edge detection stage, and discard the edge points which do not belong to a circular contour.

An illustration of the algorithm is given in Fig. 6. Fig. 6(a) shows a typical road scene with a single circular sign. Fig. 6(b) and (c) show the search region and the related edge map.

The final result of this procedure is shown in Fig. 6(f), where the successfully detected road sign is superimposed on the original image. The overall procedure, from the grey-level image to the detection of the circles takes about 15 seconds on a SparcStation ELC. Some more examples of the application of the described algorithm to different road scenes are shown in Fig. 7.

Notice that in our approach, the octagonal shape of the STOP sign is actually recovered as a quasi-circular shape. Its correct identification as an octagonal shape is, therefore, only obtained after the recognition step, where the content of the sign is analysed (see Section 3.3).

*Comparison with other algorithms*

The procedure for the recovery of circular shapes is rather slow if compared to the corresponding one for triangles. One could think of making it faster by processing edge chains instead of analysing single edge points as in the algorithm described above. Several different techniques have been proposed to extract circular-like shapes from images by making use of an analysis of the edge

[3] Recall that, under the assumptions of this paper, a circular sign does not change its shape when projected into the image.

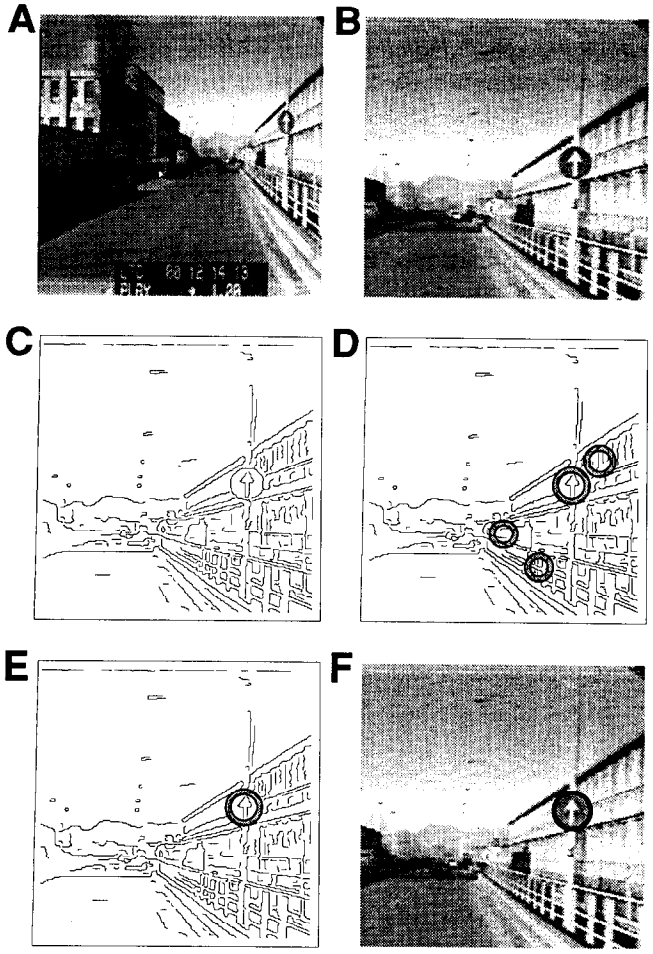



Fig. 6. Recovery of circular road signs. (a) The original image; (b) search region extracted from (a); (c) edges extracted from (b). Only the edges with a strength greater than a certain threshold are considered; (d) groups of edges obtained by matching with a set of templates are shown. The templates were 10 annuli, 3 pixels thick, with inner radius ranging from 15 to 24 pixels; the matching was simply performed by counting the edges falling inside the annuli, and by selecting the annuli with at least 30% of the surface filled by edge pixels. The resulting edges are good candidates to be parts of the contour of a circular road sign. Three tests, checking the angular and radial distribution of the edges inside the annuli, and the length of the longest edge-chain within the annulus, yield the edges shown in (e); (f) selected circular contour.

contours. Three of these have been suitably adapted to our purposes, and tested on images containing circular road signs. In this section a comparison among three algorithms (FEX by Etemadi [15], computation of local curvature [16], Hough transform [17]) and our procedure for the detection of circular shapes from images is discussed. The test images have been subdivided into two classes, roughly according to their degree of 'difficulty'. The 'easy' class contains images of scenes from highways and country roads, where the street signs stand out against a uniform background (a typical example is shown in Fig. 8(a). The 'hard' class contains images of urban road scenes where a quite cluttered background produced noisy edge maps (see for example, Fig. 8(b)) and the sign is less detectable. This difference

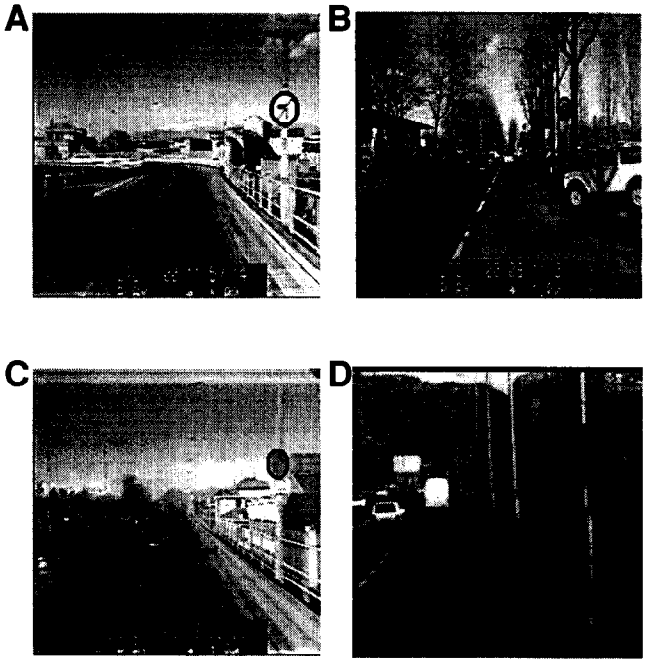



Fig. 7. Examples of circular road sign recovery. (a)–(c) Examples of successful detection; (d) example of failure, due to the lack of contrast between the circular road sign and the background.

can be appraised by looking at Figs. 8(c) and (d), where the edges extracted from Fig. 8(a) and (b), respectively, are displayed. The boundary of the street sign in Fig. 8(d) is quite deteriorated, being formed by several connected

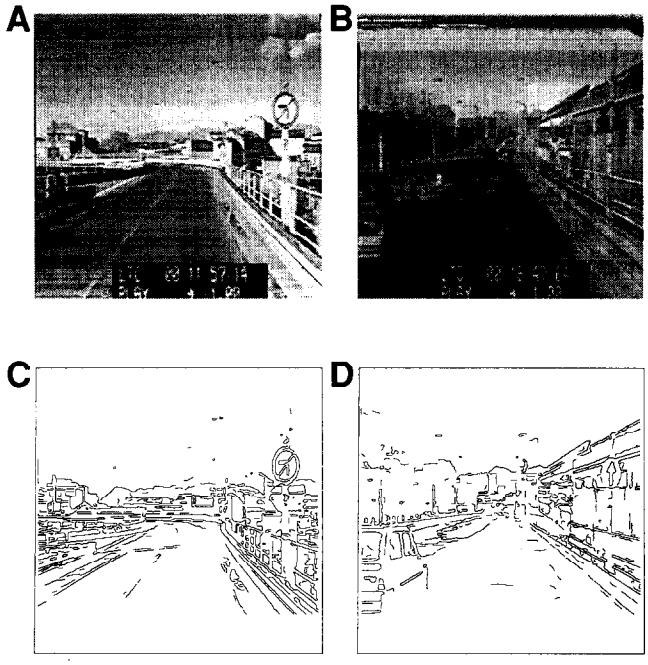



Fig. 8. An example of two different situations in road signs detection. (a) A road sign well detached from the uniform background; (b) a road sign in front of a more cluttered background; (c), (d) edges extracted from images (a) and (b), respectively. The degradation of the edges corresponding to the boundary of the road sign, which can be noticed in (d), is the reason why most of the edge-chain based methods fail in recovering curved contours in cluttered road scenes.

edge chains which are, in their turn, linked to other objects in the scene. Our procedure on the one hand is computationally more expansive than local curvature and Hough transform algorithms, but on the other hand it is able to handle 'hard' and well as 'easy' images.

**FEX.** This algorithm [15] works on edge contours, which are approximated by means of straight segments or circular arcs according to a three pass procedure which can be shortly described as follows. On the first pass the edge contours are broken into subsets which are near-symmetric about an axis passing through the mid point of the subset, and in a direction perpendicular to the line joining its end points. On the second pass these subsets are linked to form co-curvilinear and collinear segments. Finally, these segments are classified as straight-lines or circular arcs based on a simple criterion.

As an example, Figs. 9(a) and (b) show the results of such segmentation for the edge maps of Figs. 8(a) and (b), respectively.

In our application, the arcs extracted are processed to recover contours which, because of their size and shape, are apt to represent the boundary of a road sign. In spite of the good results of this algorithm on a number of synthetic and real edge maps referred to in [15], the edges extracted from images of outdoor scenes often lead to a segmentation, which does not suit our purposes, because of the strong sensitivity of the algorithm upon the edge-linking stage. This problem comes to light when the image is corrupted and cluttered: see, for example, Fig. 9(b), where the arcs representing the boundary of the sign have been lost. On the contrary, the segmentation shown in Fig. 9(a), which comes from a good quality image (Fig. 8(a)), leads to the correct identification of the sign.

**Local curvature.** Another way to obtain a similar segmentation of the edge data is to resort to a rough calculation of the local curvature of edge chains and to break the chain where the curvature significantly changes. This is achieved by a polygonal approximation of the contours [16]. Fig. 9(c) and (d) show the result of processing the images of Fig. 8 by means of the algorithm described in [16]. This procedure, originally designed for indoor image analysis, aims at recovering elliptical contours from the polygonal approximation of the edge chains. Even in this case, an extensive experimentation on real images reveals a critical dependence of the results on the accuracy of both the edge detection and the polygonal approximation process. Fig. 9(c) shows the correct localization of the sign of Fig. 8(a), whereas Fig. 9(d) shows how the true sign of Fig. 8(b) has been lost and a false sign has appeared. The usually small signal-to-noise ratio of outdoor images makes this algorithm unsuitable for road scene analysis.

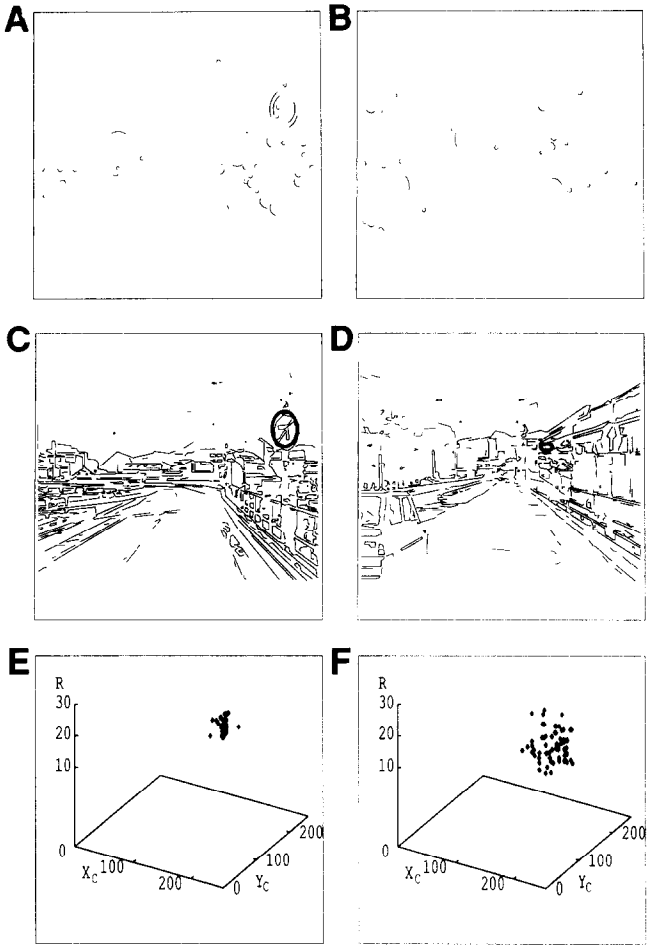



Fig. 9. Comparison among the results obtained by three different methods for the reconstruction of quasi-circular contours. The main result is that, on images like that shown in Fig. 8(b), all three methods yield a very small rate of successful detections, while on sharper images, like in Fig. 8(a), the road signs are usually detected correctly. The FEX algorithm [15] provides a segmentation of the edge data in circular arcs and straight segments. (a) (b) Circular arcs extracted by FEX from the edges of Figs. 8(c) and (d), respectively; (b) shows that the circular arcs corresponding to the road sign of Fig. 8(d) are not correctly identified. This is due to the FEX's local estimate of the curvature, which is unable to discriminate small curvature variations over short edge chains. The second method [16] has been designed to recover elliptical contours from the polygonal approximation of the edges of indoor scenes; (c) (d), respectively, show the segments obtained from Fig. 8(c) and (d), along with the recovered elliptical contours. In (c) the road sign is correctly recovered, whereas in (d) it is not. The reason is that some segments of the border of the circular sign come out to be linked to segments belonging to the background. This misleads the algorithm in the segment-merging stage. The third procedure we tested consists of computing the Hough transform of the edges. For each edge-point of the search region, the circumference through the current edge and two other edges lying 5 pixels away on the same contour is computed. In (e) and (f) the three-dimensional Hough space of the parameters of the circumferences are shown. Circumferences have been parametrized by the coordinates of the center $(x_c, y_c)$, and by the radius $R$. Only the circumferences whose centers fall within the search region and with $10 \leq R \leq 30$ pixels are shown. (e) Shows the points corresponding to the circumferences obtained from the edges of Fig. 8(c). The cluster of points clearly points at the presence of a circumference whose centre nearly coincides with the location of the road sign centre, but with a quite large uncertainty on the radius; (f) representative points corresponding to Fig. 8(d), which are too scattered to allow a confident location of the order of the street sign.

Table 1
Comparison among four different methods for the recovery of quasi-circular contours. The four algorithms have been tried on 'easy' images (200 test images with high contrast between the road sign and the background; see text and Fig. 8(a)) as well as 'hard' images (200 test images with the road sign in a cluttered background and with a consequent deterioration of the boundary of the sign; see text and Fig. 8(b)). The four rightmost columns report the results obtained by using the algorithm described in [25], the algorithm described in [13], the Hough transform of the edge contours, and the algorithm described in Section 3.2.2 respectively. In the first row, the percentage of road signs correctly detected and located is shown, while the second row gives the number of false alarms, that is reports of nonexisting road signs. The last row reports the average processing times for the four algorithms. The input data were the edges extracted from the search region

| | | Algorithm 1 | Algorithm 2 | Algorithm 3 | Algorithm 4 |
|---|---|---|---|---|---|
| Easy images | Correct detections | 87% | 98% | 95% | 96% |
| Easy images | False detections | 11 | 31 | 8 | 25 |
| Difficult images | Correct detections | 35% | 21% | 40% | 93% |
| Difficult images | False detections | 15 | 58 | 12 | 27 |
| Average processing time | | 21 sec. | 6 sec. | 8 sec. | 15 sec. |

**Hough transform.** In the Hough transform framework [17], recovering circular shapes means finding clusters of points in the 3D space of the parameters (the 2D coordinates of the centre and the radius), representing the circles with a radius within a certain range. Figs. 9(e) and (f) show the three-dimensional Hough space of parameters representing the circumferences with centre $(x_c, y_c)$ and radius $R$. For each edge point of the search region, the Hough transform is computed by recovering the circumference passing through the current point and two other edge points that lie at a certain distance on the same chain. The circumference whose centres $(x_c, y_c)$ fall outside the search region and with radius $R$ outside a certain range are discarded.

In Fig. 9(e) the cluster of representative points clearly indicates the presence of a circle whose location and average radius matches the real sign of Fig. 8(a). On the contrary, the scattering of the representative points in Fig. 9(f) does not allow the identification of a unique circumference corresponding to the true road sign.

The results of the comparison among the four algorithms (FEX, local curvature, Hough transform and ours) on a large set of images are summarized in Table 1. As explained at the beginning of this section, test images have been subdivided into two classes, the 'easy' class and the 'hard' class, according to how much the images are noisy and cluttered.

As can be seen from the first two rows of Table 1, for the 'easy' class the percentage of correct detections, as well as the number of false alarms, is comparable among the different algorithms. On the other hand, for the 'hard' class the results of the first three algorithms are quite unsatisfactory, the percentage of successful detections being too small for our purposes. Therefore, it can be concluded that these algorithms are not robust enough to recover circular contours of street signs from images of outdoor road scenes. The rightmost column of Table 1 shows that the algorithm proposed in Section 3.2.2 yields good results even in the presence of complex road environments and noisy data. Finally, the bottom row of Table 1 reports the average processing time for each algorithm.

### 3.3. Recognition of road signs

Once one or more candidates to be a road sign are found by the detection algorithm described in the previous section, a recognition stage must validate and possibly classify each candidate according to its similarity to one of the signs of a suitable database. The choice of a recognition scheme was informed by three observations. First, the output of the detection algorithm consists of potential road signs with approximately constant orientation. Therefore, no rotation of the unknown pattern is needed to match the corresponding template of the database. Secondly, the resolution and lighting of the road signs can vary considerably. Thus, the system has to cope with a large variety of realizations of each street sign in a robust way. Thirdly, the recognition procedure should present a good discriminative power at a low computational cost.

In the light of these observations, a recognition scheme based on a similarity measure between the grey-level image of the unknown road sign and the objects of the database has been adopted [18].

Fig. 10 displays the result of our algorithm on two test images which contain a STOP sign (Fig. 10(a)) and a no admittance sign (Fig. 10(b)), respectively.

The algorithm can be outlined as follows. First, the road sign is classified according to its outer shape, and then normalized to a size of $50 \times 50$ pixels by linear interpolation of the grey levels. Secondly, the normalized cross-correlation [18] between the road sign and the templates of the database related to the proper shape (see Fig. 10(e) for the case of circular signs) is computed. The best $N$ templates of the score list that fall within a fixed range of the maximum value are then considered (in both Figs. 10(c) and (d), $N = 2$, i.e. the sign #2 and #34 of the database). To increase the robustness against incertitudes due to similarities, a temporal integration of the

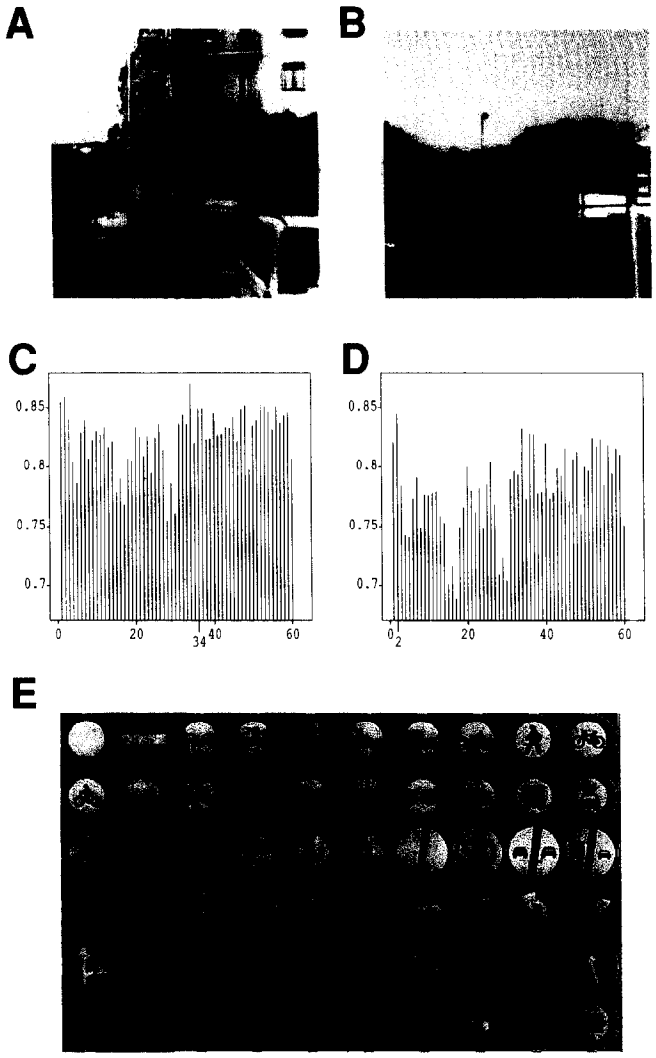



Fig. 10. Recognition of circular road signs. (a) (b) Two images, acquired in a city street, containing a STOP sign, and a no admittance sign, respectively. The circular contours detected according to the algorithm described in Section 3.2.2 are shown; (c) (d) normalized cross-correlation between the objects of the database shown in (e) and the road signs in (a) and (b), respectively. The correlation values are plotted against the ID number of the objects of the database (numbered left-right and top-down). In both cases, the maximum value of the correlation corresponds to the correct road sign (#34 for image (a) and #2 for image (b)). The distribution of the correlation values shows a similarity between the two signs. The ambiguity between the two signs can be solved by the temporal integration of the information obtained from a single image.

information obtained from a single frame can be performed as described in the following section.

An extensive experimentation has revealed a high rate of correct classifications (about 98%) and a good discriminative power even in the presence of small road signs (until $25 \times 25$ pixels wide). 'Nonsense' road signs possibly produced as a result of the detection stage, are effectively rejected by simply thresholding the value of the normalized cross-correlation. The average processing time is about 500 ms on a SparcStation ELC with a database of 60 circular signs and 47 triangular signs.

## 4. Temporal integration

The detection and recognition approach described in the previous sections has been shown to be robust against noise. Such robustness can be further increased, in the detection phase (Section 4.2), by a temporal integration based on Kalman–Bucy filtering techniques of the information obtained from a single frame by the algorithm described in the previous section. In the recognition phase, the temporal integration of the acquired information was achieved by a more straightforward method, by computing the average of the correlation between the detected sign and the signs in the database over several frames. In this case, too, the robustness of recognition was increased.

### *4.1. Kalman–Bucy filtering: generalities*

We will not give here a complete introduction to Kalman–Bucy filtering [19] and related temporal integration techniques, which have the widest use in machine vision. Suffice is to give the fundamental equations that describe the behaviour of the filter.

We have used the deterministic counterpart of the Kalman–Bucy filtering, which may be called 'recursive least-squares estimation', and which is well described in [20]. It leads to the same set of equations as its probabilistic analog, but does not require arbitrary assumptions on the probability density functions of the errors. In the following, all errors are therefore to be interpreted as maximum uncertainties.

The filter provides step-by-step estimations of the state of a linear dynamical system whose discrete-time evolution can be modelized by the process equation

$$\phi_n = T_{n,n-1}\phi_{n-1} + \epsilon_n \tag{1}$$

where $\phi_n$ is the state vector, $T_{n,n-1}$ is the evolution matrix and $\epsilon_n$ is the error on $\phi_n$ due to the error on the modelization (i.e. errors on $T_{n,n-1}$ and on the linearity hypothesis which is understood in the modelization).

Vector $\phi_n$ is not, in general, directly observed; in its place the vector $\zeta_n$ is observed, which is related to $\phi_n$ by the measurement equation

$$\zeta_n = H_n\phi_n + \eta_n \tag{2}$$

where $H_n$ is the transformation matrix and $\eta_n$ is the measurement error. The core of the filtering method is that at each time $n$ the state vector $\phi_n$ can be best estimated as the vector $\hat{\phi}_n$ recursively expressed as:

$$\hat{\phi}_i = T_{i,i-1}\hat{\phi}_{i-1} + K_i(\zeta_i - H_i T_{i,i-1}\hat{\phi}_{i-1}) \tag{3}$$

where

$$K_i = P_i^- H_i^T (H_i, P_i^- H_i^T + R_i)^{-1} \tag{4}$$

$$P_i^- = T_{i,i-1}P_{i-1}^+ T_{i,i-1}^T + Q_{i-1} \tag{5}$$

$$P_i^+ = P_i^- - K_i H_i P_i^- \tag{6}$$

and $R_i$ and $Q_i$ are the covariance matrices of the error on measurement and on model evolution, respectively. The initial condition is obtained by posing

$$P_{-1}^{+} = M_0 \tag{7}$$

where $M_0$ is the covariance matrix of the estimation of the initial state. The values of $P_0^-$, $K_0$ and $P_0^+$ can then be computed (in this order), and the recursion can start.

A simple interpretation of Eq. (3) is as follows: $K_i$ (the so-called Kalman gain) weights how much we believe in the prediction and how much we believe in the measurement. In the simple case where the state $\phi_i$ is measured directly (i.e. $H_i$ is the identity matrix), one can be convinced of this by observing that if $K_i$ is 0, $\hat{\phi}_i$ coincides with the prediction $T_{i,i-1}\hat{\phi}_{i-1}$, whereas if $K_i$ is 1 (more properly, the identity matrix), $\hat{\phi}_i$ coincides with the measure $\zeta_i = \phi_i^{(\text{measured})}$

### 4.2. Tracking the road sign

Let us now see how all this applies to our problem, which is to obtain a reliable tracking of road signs. The vehicle is assumed to be moving with approximately constant speed $\vec{v} = (0, 0, v)$ in the $Z$-direction, on an approximately straight road with constant slope. Corrections to the assumed straight and uniform motion will be implemented in the state equation as errors on the $\vec{X}(t)$ of the road sign (e.g. the centre of the circle for a forbidding sign), the speed $v = \dot{Z}$ of the moving vehicle, the real size $\Sigma$ of the sign (which is equal to the diameter of the circle in the case of a forbidding sign, and to one side of the triangle in the case of a warning sign). The measured quantities are the location $(x_C(t), y_C(t))$ of the detected road sign in the image and its apparent size $\sigma(t)$.

We want to track the centre of the sign, defined by the following equation, which relates the essential state variables to the measured variables:

$$(x_C(t), y_C(t)) = f\left(\frac{X_0}{Z_0 + vt}, \frac{Y_0}{Z_0 + vt}\right) \tag{8}$$

where $(X_0, Y_0, Z_0)$ is the location of the road sign at time $t = 0$; in our model, the quantities $X_0$ and $Y_0$ are assumed not to change.

The apparent size $\sigma(t)$ is related to the real size $\Sigma$ by the following equation:

$$\sigma(t) = \frac{f\Sigma}{Z_0 + vt} \tag{9}$$

The evolution of the system is considered only at discrete times $t_n$. The time interval $\Delta t_n = t_n - t_{n-1}$ between two different states is not necessarily constant.

The equation of the state $\Phi_n = (X_n, Y_n, Z_n, \dot{Z}_n)$ describing the location of the road sign can be written as

$$\begin{cases} X_n = X_{n-1} + \epsilon_{X,n} \\ Y_n = Y_{n-1} + \epsilon_{Y,n} \\ Z_n = Z_{n-1} + \dot{Z}_{n-1}\Delta t_n + \epsilon_{Z,n} \\ \dot{Z}_n = \dot{Z}_{n-1} + \epsilon_{\dot{Z},n} \end{cases} \tag{10}$$

which account for a motion along the $Z$-axis at a constant speed. $\epsilon_{X,n}$, $\epsilon_{Y,n}$, $\epsilon_{Z,n}$ and $\epsilon_{\dot{Z},n}$ are the errors on the model. $\epsilon_{X,n}$, $\epsilon_{Y,n}$ and $\epsilon_{\dot{Z},n}$ are assumed to be independent and constant errors, respectively equal to $\delta X$, $\delta Y$, $\delta v$. The main contribution to $\epsilon_{Z,n}$, which is also constant, is assumed to be the error $\delta v$ on the estimated speed $\dot{Z}_n$, $\delta v$, so that $\epsilon_{Z,n} = \delta v \Delta t_n$.

The evolution matrix is

$$T_{n,n-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t_n \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{11}$$

The covariance matrix of the errors on the model can be approximated (by considering the contributions $\delta X$, $\delta Y$, $\delta v$ to be independent) as

$$Q_n = \begin{pmatrix} (\delta X)^2 & 0 & 0 & 0 \\ 0 & (\delta Y)^2 & 0 & 0 \\ 0 & 0 & (\delta v \Delta t_n)^2 & (\delta v)^2 \Delta t_n \\ 0 & 0 & (\delta v)^2 \Delta t_n & (\delta v)^2 \end{pmatrix} \tag{12}$$

Apart from the uncertainty in the motion model, which is reflected by the terms containing $\delta v$, we have added diagonal terms to $Q_n$ to avoid a very fast decrease in the state covariance.

The measurement equations are simply

$$\begin{cases} x_n = f\dfrac{X_n}{Z_n} + \eta_{x,n} \\ y_n = f\dfrac{Y_n}{Z_n} + \eta_{y,n} \\ \sigma_n = f\dfrac{\Sigma_n}{Z_n} + \eta_{\sigma,n} \end{cases} \tag{13}$$

where $\eta_{x,n}$, $\eta_{y,n}$ and $\eta_{\sigma,n}$ are independent errors of equal value $\delta u = 2$ pxl. Observe that the measurement equations are not linear in the state variables, so that the Kalman-filter framework cannot be adopted straightforwardly. An extended Kalman filter approach needs to be adopted by linearizing the measurement equation around the predicted state [20].

Let us define

$$\hat{\Phi}_n^- = T_{n,n-1}\hat{\Phi}_{n-1} \tag{14}$$

where $\hat{\Phi}_{n-1}$ is the Kalman state at time $t_{n-1}$.

By introducing the variables

$$x'_{C,n} = x_{C,n} - f\frac{\hat{X}_n^-}{\hat{Z}_n^-}$$

$$y'_{C,n} = y_{C,n} - f\frac{\hat{Y}_n^-}{\hat{Z}_n^-} \tag{15}$$

$$\sigma'_n = \sigma_n - 2f\frac{\Sigma}{\hat{Z}_n^-}$$

the new measurement equations become

$$\begin{pmatrix} x'_{C,n} \\ y'_{C,n} \\ \sigma'_n \end{pmatrix} = H_n \begin{pmatrix} X_n - \hat{X}_n^- \\ Y_n - \hat{Y}_n^- \\ Z_n - \hat{Z}_n^- \\ \dot{Z}_n - \hat{\dot{Z}}_n^- \end{pmatrix} + \begin{pmatrix} \eta_{x,n} \\ \eta_{y,n} \\ \eta_{\sigma,n} \end{pmatrix} \tag{16}$$

where

$$H_n = \begin{pmatrix} \frac{f}{\hat{Z}_n^-} & 0 & -f\frac{\hat{X}_n^-}{(\hat{Z}_n^-)^2} & 0 \\ 0 & \frac{f}{\hat{Z}_n^-} & -f\frac{\hat{Y}_n^-}{(\hat{Z}_n^-)^2} & 0 \\ 0 & 0 & -f\frac{\Sigma}{(\hat{Z}_n^-)^2} & 0 \end{pmatrix} \tag{17}$$

The covariance matrix for the measurement error, $R_n$, is simply

$$R_n = (\delta u)^2 \,\mathrm{diag}(1,1,1) \tag{18}$$

The initial state can be found by estimating $X_0$, $Y_0$, $Z_0$ and $v_0$. In practice, the 'location' variables $X_0$, $Y_0$, $Z_0$ can be estimated by means of an automatic or manual estimation of $(x_0, y_0, \sigma_0)$ and by a subsequent back-projection to obtain the 3D quantities:

$$(X_0, Y_0, Z_0) = \frac{\Sigma}{\sigma_0}(x_0, y_0, f) \tag{19}$$

The error on such estimates essentially depends on the errors $\delta u_x$, $\delta u_y$, $\delta u_\sigma$ on the 2D quantities $x_0, y_0, \sigma_0$, whereas the estimation of the speed is given a priori and is rather wide. The error on the estimation of $\Sigma$ and $f$ are comparatively low and can be neglected:

$$\begin{cases} \delta X_0 = X_0\left(\frac{\delta u_x}{x_0} + \frac{\delta u_\sigma}{\sigma_0}\right) \\ \delta Y_0 = Y_0\left(\frac{\delta u_y}{y_0} + \frac{\delta u_\sigma}{\sigma_0}\right) \\ \delta Z_0 = \Sigma\left(\frac{\delta u_\sigma}{\sigma_0}\right) \\ \delta \dot{Z}_0 = \delta v \end{cases} \tag{20}$$

The errors $\delta u_x, \delta u_y, \delta u_\sigma, \delta v$ are assumed to be independent. It is also assumed that $\delta u \equiv \delta u_x = \delta u_y = \delta u_\sigma$. The covariance matrix of the initial state is thus given by

$$M_0 = \begin{pmatrix} X_0^2\left(\frac{1}{x_0^2} + \frac{1}{\sigma_0^2}\right)(\delta u)^2 & \frac{X_0 Y_0}{\sigma_0^2}(\delta u)^2 & \frac{X_0\Sigma}{\sigma_0^2}(\delta u)^2 & 0 \\ \frac{X_0 Y_0}{\sigma_0^2}(\delta u)^2 & Y_0^2\left(\frac{1}{y_0^2} + \frac{1}{\sigma_0^2}\right)(\delta u)^2 & \frac{Y_0\Sigma}{\sigma_0^2}(\delta u)^2 & 0 \\ \frac{X_0\Sigma}{\sigma_0^2}(\delta u)^2 & \frac{Y_0\Sigma}{\sigma_0^2}(\delta u)^2 & \frac{\Sigma^2}{\sigma_0^2}(\delta u)^2 & 0 \\ 0 & 0 & 0 & (\delta v)^2 \end{pmatrix} \tag{21}$$

Now that all the essential quantities and equations have been defined, let us take a closer look at how the filtering scheme is implemented in practice.

After the initialization step, in which the presence of the road sign is investigated in all the search region (estimated as described in Section 3.1), the road sign is tracked until it disappears from the image. The tracking allows a drastic reduction of the size of the search region with respect to the single-image case. In fact, if the 2D location of the sign in image $\mathcal{I}_{i-1}$ is known, the search region in image $\mathcal{I}_i$ will be centred in the predicted 2D location of the sign and its size will be given by the predicted 2D size of the sign augmented with the expected errors on location and size (see also [21]).

More specifically, the search region will be a box of size

$$\Delta x = \sigma_n + \sqrt{(H_n P_n^- H_n^T + R_n)_{x,x}} + \sqrt{(H_n P_n^- H_n^T + R_n)_{\sigma,\sigma}} \tag{22}$$

$$\Delta y = \sigma_n + \sqrt{(H_n P_n^- H_n^T + R_n)_{y,y}} + \sqrt{(H_n P_n^- H_n^T + R_n)_{\sigma,\sigma}} \tag{23}$$

A delicate point in the tracking process, which is related to the computation of the expected size of the search region, is the choice of the time interval between a measure and the following, $\Delta t_n$. In our scheme, a tentative measure is carried out at each frame, and if the road sign is always detected $\Delta t_n$ is constant and equal to the interval between two successive frames; otherwise, $\Delta t_n$ is the time between two successful measures, that is between two frames in which the sign has actually been detected. Since we do not know a priori when a successful measure will take place, however, the matrices $H_n$, $P_n^-$, $P_n^+$ and $K_n$ must be computed at each frame. If no sign is detected, the measure is aborted, and all matrices are computed again with an increased time interval until a successful measure is carried out.

An example of tracking of a circular sign in a 20 frames sequence is shown in Fig. 11. The sign is first detected when its apparent size reaches 10 pxl. Each time the sign is detected its apparent contour is superposed onto the image. In Fig. 12, the Kalman equation of the $Z$-coordinate of the road sign, $\hat{Z}_n$ is plotted against

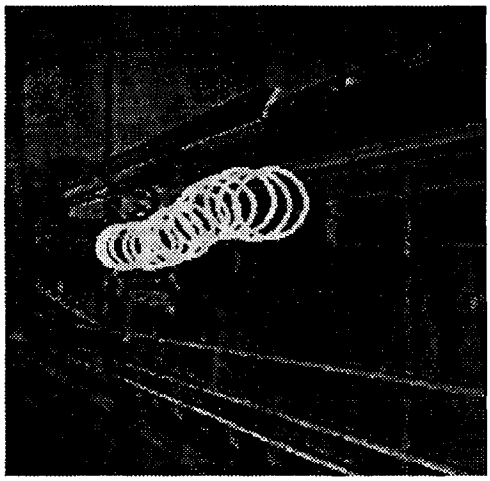


Fig. 11. The first frame of a 20 frames sequence, in which a road sign appears whose initial apparent size is 8 pxl. The road sign is first detected when the apparent size reaches 10 pxl (frame no. 4). Each time a sign is detected and recognized its apparent contour is superimposed onto the original image. Observe that, due to changes in the direction of the optical axis of the camera with respect to the road direction, the centre of the sign does not move exactly on a straight line, as predicted by the evolution model.

time. The graph of $\hat{Z}_n$ (dotted line) is compared to the step-by-step prediction of $Z_n$, $T_{n,n-1}\hat{Z}_{n-1}$ (dashed line), and to the 'measured' $Z_{n,\text{meas}}$ (plain line). $Z_{n,\text{meas}}$ is not measured directly, but through the apparent size $\sigma(t)$ of the road sign, which is related to $Z_{n,\text{meas}}$ by the equation

$$Z_{n,\text{meas}} = f\frac{\Sigma}{\sigma_n} \tag{24}$$

The graphs of the measured and predicted $x$- and

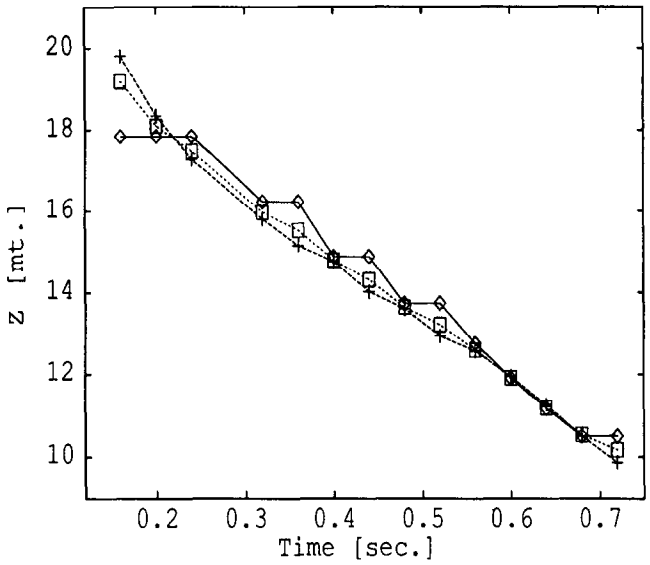



Fig. 12. The result of the temporal integration. The graph of the $Z$-coordinate of the Kalman state is plotted against time and compared to the graph of the predicted and 'measured' state.

$y$-coordinates of the road sign in the image are shown in Fig. 13.

### 4.3. Temporal integration of recognition

We have not found it necessary to address the recognition problem by Kalman methods, as a simple weighted time average of the cross-correlation of the image road sign with the database signs gives very good results.

The recognition output at frame $i$ is simply an $N$-tuple of cross-correlation values $C(i,n)$ $(i-1,\ldots N)$ of image $i$

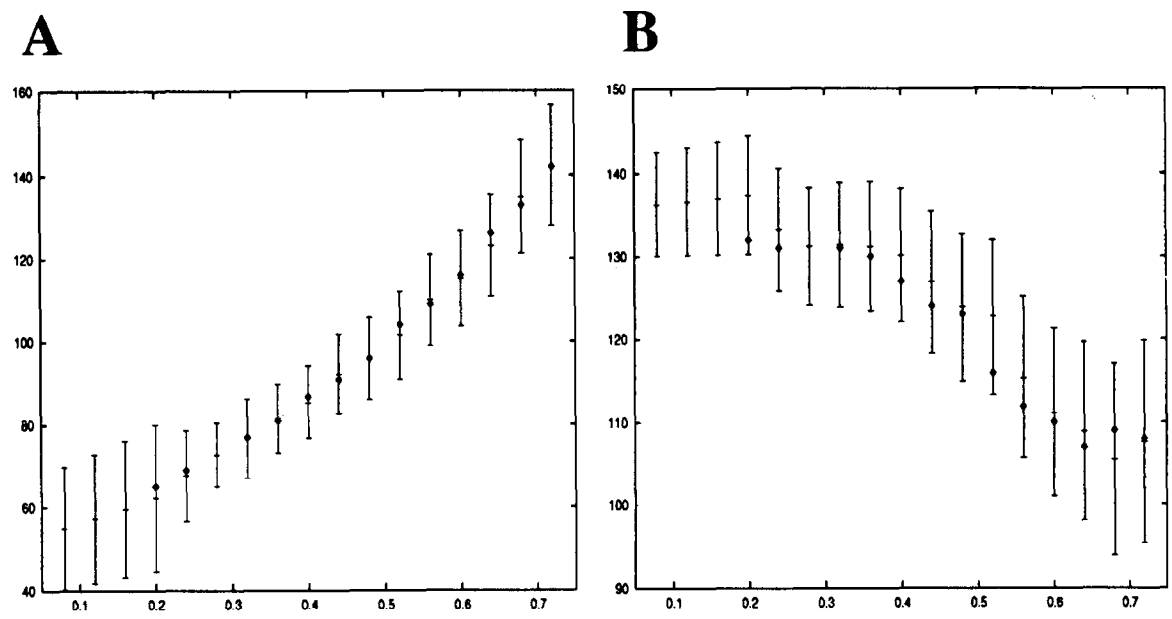



Fig. 13. The measured (+) and predicted (◇) $x$- and $y$-coordinates of the road sign in the image, plotted against time. The predicted values are given with their error bars.

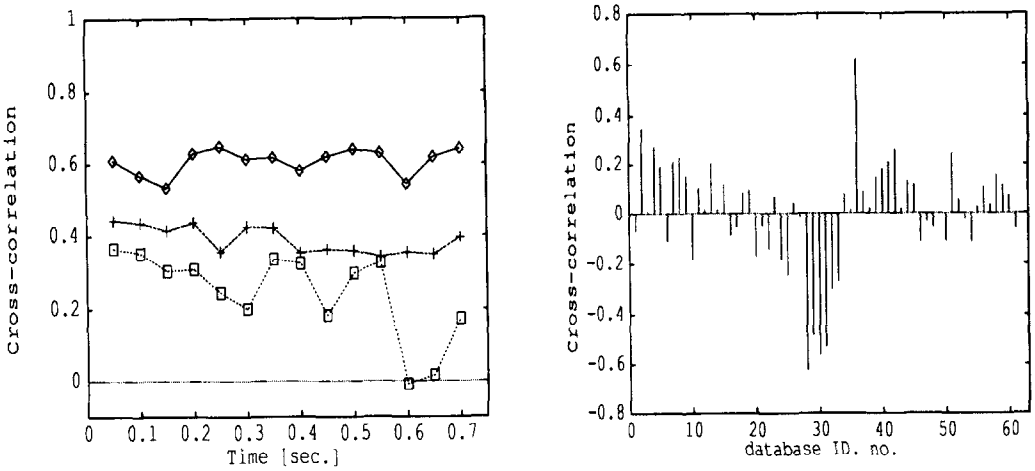



Fig. 14. (a) The weighted sum of the cross-correlations between the detected sign in the image and the signs in the database which are the best matches throughout the sequence. As can be seen, the best match tends to stand out steadily with respect to the other signs in the long run, virtually reducing to zero the number of wrong identifications; (b) weighted sum of the cross-correlations between the detected sign in the image and all the signs in the database. The correct sign (#36) strongly stands out.

with the $n$th sign of the database. If the sign has been detected in $K$ images, an average cross-correlation $C^{(K)}(n)$, which takes into account all cross-correlations and weights them according to the radii of the detected sign, is defined:

$$C^{(K)}(n) = \frac{\sum_{i=1}^{K} C(i,n)\sigma_i^2}{\sum_{i=1}^{K} \sigma_i^2} \tag{25}$$

where $\sigma_i$ is the size of the detected sign at image $i$. As an example, let us analyse the sequence to which we already referred in the previous section. In Fig. 14(a), where $C^{(K)}(n)$ is plotted against $K$ for those elements of the database whose correlation was almost always close enough to the 'true' sign, the graph of the true sign tends to stand out steadily as time goes by.

For the sake of completeness, the value of $C^{(K)}(n)$ after $K = 14$ successful recognitions for all the elements $n = 1, \ldots N$ of the database is shown in Fig. 14(b).

## 5. Discussion

In this paper a novel approach to the detection and recognition of road signs has been presented. A simple analysis of the edges extracted from monochromatic or colour images, along with an approximate a priori knowledge on the scenario, makes it possible to detect and recognize the road signs even in quite complex scenes. Special attention has been devoted to the robustness of the whole system against variations of the lighting conditions and the poor accuracy of the low-level processing; an additional gain in robustness has been achieved by the temporal integration of the information extracted from the analysis of a single frame. The reliability demonstrated by the proposed method in many different conditions suggests that this module could be part of an integrated driver assistant system based on computer vision technology.

### 5.1. Relation to previous work

Several different techniques have been proposed for the detection of road signs from sequences of monochromatic [10] or colour images [11–13,16,22].

In [10] triangular, octagonal and circular contours, which are likely to represent boundaries of road signs, are selected only among closed edge-contours. This makes the algorithm strongly dependent on the quality of the edge detection process. In [16] a more complicated strategy is followed, which uses both colour and edges clues; edges are examined at different levels of resolution by using what the authors call a Hierarchical Structure Code, which allows to pass from the signal space of an image into the space of its symbolic representation; it is assumed that closed edge-contours are available at one of these levels of resolution, and failures happen when the outline of the traffic sign merges with the background.

A scheme for shape recognition, based on uncertainty handling and multiple knowledge sources combination and propagation, has been applied to correctly segmented images [9]. Tree structure model based systems indexed by shape, colour and pictogram features have been implemented for the recognition of the detected road signs [12].

Janssen et al. [11] developed a detection and recognition scheme for road signs which also subdivides the process into three natural stages: (i) extraction of a search region, (ii) shape detection, and (iii) recognition. The rationale of the approach, however, is different. Unlike [11], who rely on classifiers which identify shapes and pictograms after a learning stage, all our routines are

based on low-level techniques such as edge detection, polygonal approximation and template matching, with bits of simple and robust geometrical reasoning for the detection of shape. Furthermore, our approach is essentially grey-level based, although colour is used – when available – to reduce the search region. Colour is not considered, in our work, as a reliable tool to produce candidates for the recognition step, that is to produce regions which exactly bound road signs. Experience has shown us, indeed, that in corrupted images colour-connected components that exactly correspond to road signs can hardly be isolated, no matter how thresholds for the hue, saturation and value (brightness) of the image are chosen; this is also true if the image contains many objects of the same colour as the road sign, which often happens in urban streets where houses and cars are present. Heuristics of the kind developed in [11,12] with an access language which investigates topological relationships between uniformly coloured regions is also very useful, but can fail under particular weather and lighting conditions (e.g. in the case of Fig. 3, which shows a photograph taken at sunset so that there is a strong red component throughout the image, the white region inside the red border of the circular sign is identified as red by our algorithm) and when, on account to disconnections in the colour-connected components due to noise, the topological relationships between different components are violated.

We have thus found it more practical to use colour (when available) only in the extraction of the search region, thenceforth sticking to the grey-level information on edges for the shape detection stage. We do not count on reliable edges, because they soldem occur in real outdoor images. The methods that we have implemented have proven to be stable against noise, as the geometrical analysis of edges that we carried out does not require that the edges be of good quality (see Fig. 8). Observe that processing image edges is much faster than methods of shape detection based on template-matching applied to luminance and chrominance image data.

Apart from its practical relevance, this work also has a methodological utility, in that it shows that an approach to shape detection based on a geometrical analysis of contours is indeed possible, even in the delicate context of cluttered road scenes. This approach has been criticized, on good grounds, by several researchers (see, for example, Janssen et al. [11]), who pointed at the inherent unreliability of the edge detection stage. Our work shows, however, that it is not the resort of low-level techniques such as edge detection and linking which should be in debate, but the adoption of a purely bottom-up paradigm, according to which one should extract linked contours of edges and then inquire if one of the extracted contours has the desired shape. This approach, which is doomed to failure in case of noisy and cluttered data, must be replaced with a model-based approach, according to which we expect sets of points of given shape and size and we search for them in the image in ways that are designed *ad hoc* for the environment. In this framework, low-level techniques only serve at the purpose of generating hypotheses (i.e. candidates to be road signs) that must be subsequently tested by a recognition stage.

The same approach will be used in the future for the construction of a system which is able to recover the 3D information on the road scene by the visual information obtained by a single camera mounted on a moving vehicle. This work is intended to be part of such a system.

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